

PARITY — review cheat sheet

Adaptive horror driven by online player-trajectory prediction · 19 August 2026

The 40-second version. Horror games die when the player learns them. Existing antagonists are static (learnable, stops being scary) or random (unlearnable, feels unfair). PARITY builds an online behavioural model of the individual player — a variable-order Markov model over their route through a room graph — and uses it to intercept where they are going and to poison the information they depend on. The model is the deliverable; the game makes it legible.

Numbers to know cold

RESULT	VALUE	THE POINT
Top-1 next-move accuracy, habitual player	60.5% vs 46.5% (order-1) vs 30.5% (none)	The model learns routes
Same, random player (<i>the control</i>)	26.7% vs 25.9% vs 26.9%	Identical — no leakage
Predicting "will they stay put?"	70% vs 54% baseline; 100% in empty rooms	It learned where you stop to write
Captures / 1000 moves	random 111 · hub-camp 265 · order-1 305 · VOMM 319	Any model beats none; models are close
World editing (no entity)	20.9 → 32.3 moves per objective lap	+54% cost , no soft-locks
Advantage over hub-camping vs facility size	1.15× at 10 rooms → 1.57× at 32	Prediction matters more at scale
Evasive player (declines 35% of windows)	capture risk −1.3%; model accuracy −4.6 pts	Negative result — motivates M6

10 rooms, 1200 windows × 12 seeds, prequential. 9 invariants pass over 25 generated facilities.

Questions you will get

What is a VOMM?

Variable-order Markov model. It predicts your next move from your recent path, using as much history as it has reliable data for and backing off to less when it doesn't. Order-1 can't represent routes, because a route means the same room has different successors depending on how you entered it. Fixed high order can, but never has enough data. Variable order gets both.

Isn't this just dynamic difficulty adjustment?

No. DDA tunes difficulty *magnitude* — enemy health, encounter budget. This changes *where the antagonist chooses to be*, from a model of the individual. That gap is point 1 of the research gap, across 22 papers.

Why 61% and not higher?

Because staying put is a legal move. Predicting whether someone moves at all is harder than predicting where. An earlier version forbade waiting and scored 90% — that number was inflated against a player who couldn't defend, and I corrected it.

Hub-camping gets 265 without any model. Why bother?

Only on a small map. At 32 rooms the advantage is 1.57× and widening — the ten-room result understates prediction's value. I included hub-camping deliberately: a model that can't beat "sit in the busiest room" hasn't earned its complexity.

Can the player beat it?

Not yet, and that's my most useful result. I built a bot to defeat my own AI by waiting. It degrades the model's accuracy but barely reduces capture risk, because the antagonist currently observes position perfectly — standing still just makes you a stationary target. Next milestone senses the player through door motion-detectors instead, which makes it a partially-observed problem and makes waiting a real tactic.

How do you know it isn't cheating / overfitting?

Two ways. Evaluation is prequential — every prediction is made before its outcome is observed, so there's no train/test leakage by construction. And the control: against a random player the model scores identically to no model. If it beat the control, something would be leaking.

Where's the game?

It doesn't exist yet, and I won't claim otherwise. What exists is the simulation core, the model, the director, ten experiments and nine invariants — plus a browser demo of the model and a walkable art-direction study. The Godot build is milestone M3.

Why Markov and not deep learning?

Cold start. A horror antagonist has to be interesting in the first two minutes against a player it has never seen, with no corpus for that individual. VOMM learns online from the first move, persists across runs, and is inspectable — which is what makes the in-game readout possible. A GRU or an STRelay-style architecture is the stretch goal, behind the same interface.

Concede these fast, don't defend them. (1) The models are within noise of each other at interception — the clear win is at prediction, not capture rate. (2) Five-step lookahead is now slightly worse than one-step, because rollout error compounds at 61% accuracy. (3) Dwell rate is a modelling assumption calibrated to ~24% of windows, not measured human behaviour — that's what the playtest milestone is for. Owning these reads as rigour; being caught on them doesn't.

Demo running order

#	WHAT	SAY
1	parity-demo.html?auto=1	"The model, live." Point at the belief panel, then the accuracy comparison against the no-model control, then the learned-rule readout in plain English.
2	parity-room.html	"The look we're building toward." Walk one room. Say plainly it's an art study, not the game — no entity, no clock, no notebook.
3	python3 model/test_invariants.py	Offer it if they press on rigour. Nine invariants, a few seconds.